HIGH-RESOLUTION 3D CT SYNTHESIS FROM BIDIRECTIONAL X-RAY IMAGES USING 3D DIFFUSION MODEL

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ABSTRACT

3D Computed Tomography (CT) offers invaluable geometric insights into bone structures, but the high radiation dose and medical cost constraints are significant barriers. Moreover, CT reconstruction demands multiple X-ray projections, necessitating a dedicated scanning system, whereas bidirectional X-rays are already the front-line diagnostic tool in routine practice. Therefore, reconstructing 3D bone structures from bidirectional X-ray data can reduce the need for additional CT scans, provide rapid access to 3D information, and lower medical costs. Recently, diffusion models have emerged as potent tools for generating high-fidelity images. However, high computational costs have limited their utility. Collecting large-scale datasets for training in clinical environments presents another challenge.

In this study, we introduce a novel approach to synthesize 3D CT volumes from a bidirectional X-ray projection using a 3D diffusion model. To reduce the computational burden and the need for a large dataset, our 3D diffusion model was trained using patch-wise loss. A conditional score function of our model incorporates 2D bidirectional X-ray images and patch coordinate information to synthesize high-resolution CT. Initial findings indicate that our diffusion model synthesizes 3D CT volumes from a bidirectional X-ray, effectively capturing 3D geometric correlations while enabling single-GPU training and rapid 3D volumetric sampling.

***Index Terms*—** CT, X-ray, Deep learning, Diffusion Model

1. INTRODUCTION

Bidirectional X-ray imaging is commonly used for initial wrist fracture diagnosis. However, 3D CT scans are considered the gold standard for patient-specific implant selection and surgical planning in the preoperative phase, despite radiation exposure and cost concerns. Therefore, reconstructing 3D bone structures from bidirectional X-ray data may reduce the need for additional CT scans, provide rapid access to 3D information, and lower medical costs. However, the main challenge lies in the inherent ambiguity of 3D information within X-ray images.

Recently, deep learning technology has emerged as a promising solution to address these challenges [1, 2]. Deep learning models can be trained to generate realistic 3D bone structures directly from 2D X-ray images, leveraging large three-dimensional image data. In particular, Generative Adversarial Networks (GANs) have been applied to cross-domain image translation in medical imaging, demonstrating their effectiveness [3, 4], but GANs are known to have difficulties due to mode collapse and training instability.

Diffusion models have recently gained attention due to their excellent performance in generating realistic images while maintaining training stability [5, 6, 7]. This is because diffusion models enhance the data representation capacity of the model by learning the gradient of the log density of the data datasets. The operation of diffusion models begins with pure noise and iteratively removes noise to generate the sample of target data distribution. As a result, the generation process operates in the same dimension as the original data, which becomes problematic when extending diffusion models to multi-modal tasks like bidirectional X-ray images, as X-ray images are 2D, while the target dimension is 3D. Furthermore, during the training of 3D diffusion models, 3D data can easily exceed GPU memory limitations, leading to limited resolution and the need for large datasets [8].

In this study, we propose a 3D diffusion model to synthesize high-resolution 3D CT volumes from 2D X-ray images. Our 3D diffusion model was designed as a score-based generative model (**Fig.1**). To enable training the 3D diffusion model on a single GPU, we use the sub-regions extracted

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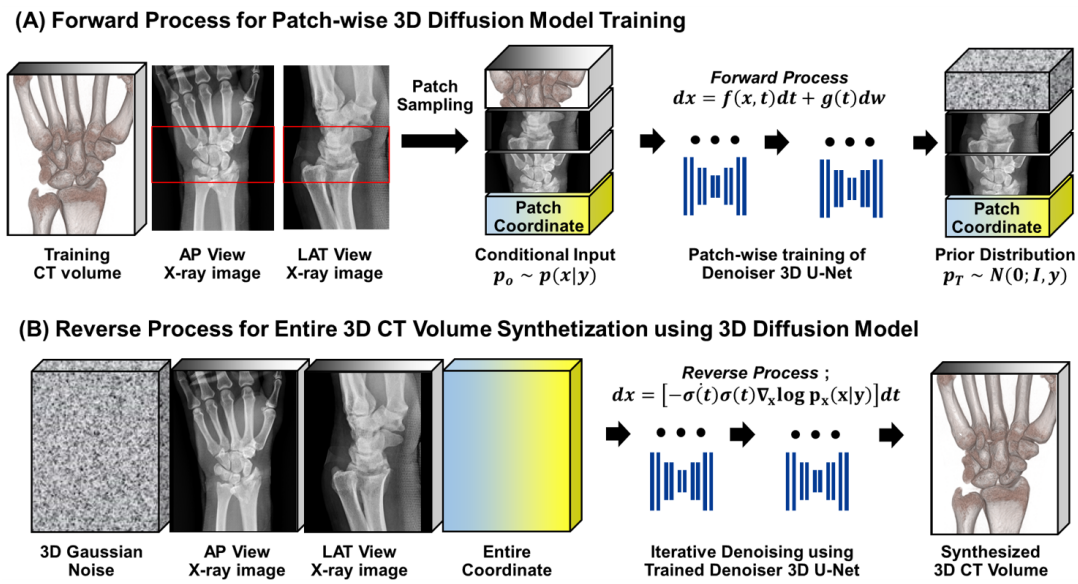

Fig. 1. The outline of the proposed 3D diffusion model; (A) The forward processing of the diffusion model involves patch-wise training of the Denoiser 3D U-Net, which takes conditional input in the form of patches consisting of the training CT volume, anterior-posterior (AP) view, and lateral (LAT) view x-ray images, along with patch coordinates. (B) To synthesize the entire 3D CT volume using the patch-wisely trained denoiser 3D U-Net, the 3D Gaussian noise and AP/LAT x-ray images are iteratively denoised, following the probability flow ordinary differential equation.

from the x-ray images in the superior and inferior direction and regional coordinates as conditional inputs to generate the corresponding 3D sub-volume. The trained model samples a high-resolution 3D CT volume (0.5 mm^3) using bidirectional 2D images and coordinates of the entire image. The proposed 3D diffusion model-based CT synthesis was trained and validated using 3D wrist CT and digitally reconstructed radiography (DRR).

2. SCORE BASED DIFFUSION MODELS

The diffusion model is a generative model consisting of two processes: forward and reverse processes. In the forward process, the conditional data distribution, $p(x|y)$, is diffused by adding Gaussian noise to generate the series of noisy samples,

$$p(x_t|x_0, y) = \mathbf{N}(x_t; \mu_t x_0, \mu_t^2 \sigma_t^2 I), \tag{1}$$

where the condition y is provided. The forward process can be interpreted as a continuous mapping from data distribution to prior distribution with time variable $t \in [0, T]$ such that $x(0) \sim p_0$, which equals a data distribution, and $x(T) \sim p_T$ equals a prior distribution. In the view of score-based generative modeling [6], the forward processing could be characterized by a stochastic differential equation (SDE),

$$dx = f(x,t)dt + g(t)dw, \tag{2}$$

, where w is Brownian motion, $f(\cdot, t) : \mathbf{R}^d \to \mathbf{R}^d$ is the drift function, and $g(t) : \mathbf{R} \to \mathbf{R}$ is a diffusion coefficient.

In the reverse process, we can sample the data distribution $x(0) \sim p_0$ started from the prior distribution $x(T) \sim p_T$, while the backward processing is also given by the reverse time SDE. The closed form of reverse time SDE is provided in Anderson [9] and Song et al.[6] as,

$$dx = [f(x,t) - g(t)^2 \nabla_x \log p_t(x|y)]dt + g(t)d\bar{w}, \tag{3}$$

where $d\bar{w}$ is a Brownian motion when time flows reversely, starting from $t = T$ to $t = 0$. Here, the gradient of the log density of the data datasets, $\nabla_x \log p_t(x|y)$, is a score function. The SDE in the reverse process, Eq. 3, can be converted into the ordinary differential equation (ODE) named probability flow ODE [10]. This is given as

$$dx = [-\dot{\sigma}(t)\sigma(t)\nabla_x \log p_t(x|y)]dt, \tag{4}$$

where $\sigma(t)$ is a change of standard deviation over time. In order to solve the reverse process, the score function $\nabla_x \log p_t(x|y)$ has to be determined. Therefore, the neural network can be trained to estimate score function by score matching with a denoiser function $D(x; \sigma, y)$,

$$\nabla_x \log p_t(x|y) = (D(x; \sigma, y) - x)/\sigma^2. \tag{5}$$

The loss function of the denoiser function is given as

$$L := \mathbf{E}_{p_{data}} \mathbf{E}_{n \sim N(0, \sigma^2 I)} \|D(x + n; \sigma, y) - x\|_2^2, \tag{6}$$

3. PATCH-WISE TRAINING OF DIFFUSION MODEL

The training of a 3D diffusion model demands significant memory and is made even more challenging by the need to

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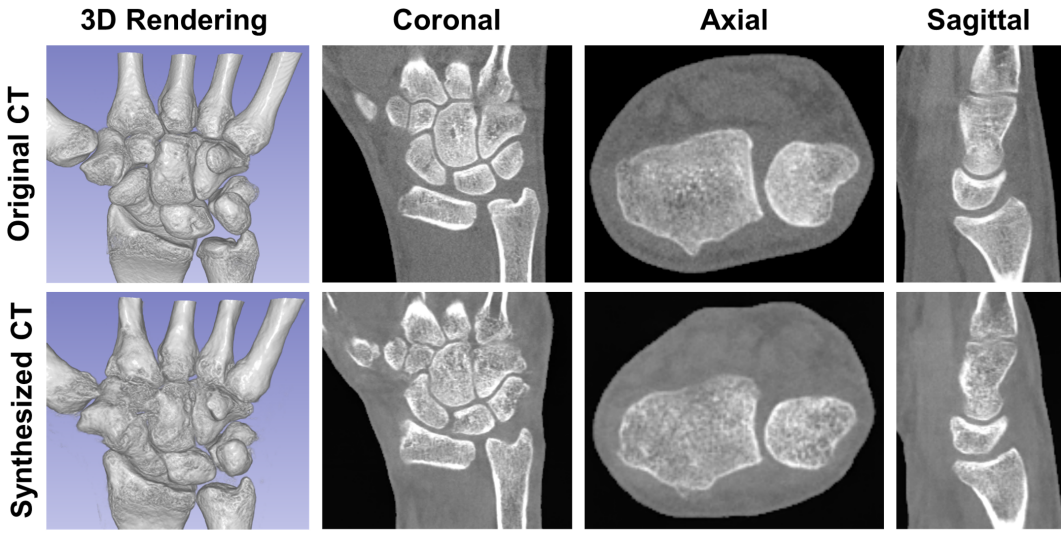

Fig. 2. Representative wrist CT images; The synthesized wrist CT (Bottom) accurately represents bone anatomy and matches the original CT (Top), as seen in both 3D volume rendering and slice images.

gather extensive 3D CT data. To reduce memory usage and the need for a large dataset, we trained a 3D diffusion model using patch-wise loss [11]. The loss function of the denoiser function can be decomposed as the sum of patch-wise losses because convolution layers are shift-invariant. We use x_Ω to denote a uniformly sampled of 3D CT sub-volume, x, then $x = \bigcup_{i=1}^{N} x_{\Omega_i}$, where N is a total number of sub-divisions. Then, the patch-wise loss is given as

$$L_P = \sum_{i=1}^{N} \mathbf{E}_{p_{data}} \mathbf{E}_{N(0,\sigma^2 I)} \| D(x_{\Omega_i} + n; \sigma, y) - x_\Omega \|_2^2. \quad (7)$$

4. EXPERIMENTS

We included a total of 76 patients, with 32 having right-hand and 44 with left-hand 3D CT scans. To prepare the data, we performed body-part segmentation to eliminate extra-body objects, such as splints or casts, using ITK-snap [12]. Additionally, the left-hand CT scans were flipped to have identical anatomical positions as the right-hand scan. The segmented 3D CT volumes have the size of $256 \times 256 \times 160$, which corresponds to a spatial resolution of 0.5 mm^3.

We randomly split the dataset into 80%/20% for model training (n=61) and testing(n=15), respectively. Our neural network is a modified 3D DDPM++ U-net model [10] by changing convolutional layers and attention modules to 3D. For the patch-wise training, a 3D CT volume was randomly sampled using a 64-slice window in the superior-inferior direction, which can be seen in both AP and LAT view images. The 3D sub-volume has the size of $256 \times 64 \times 160$. Subsequently, the 3D sub-volume was projected in the AP view (256×64) and LAT view (64×160) to create DRR patch images while preserving resolution in the superior-inferior direction.

Method	PSNR	SSIM
FBP (2 Views)	8.48 ± 0.48	0.08 ± 0.01
FBP (36 Views)	17.54 ± 0.68	0.29 ± 0.03
Diffusion (NFE=10)	16.66 ± 0.69	0.23 ± 0.02
Diffusion (NFE=50)	16.78 ± 0.82	0.42 ± 0.02
Diffusion (NFE=100)	16.51 ± 0.77	0.42 ± 0.02
Diffusion (NFE=1000)	16.71 ± 0.74	0.42 ± 0.02

Table 1. Quantitative Evaluation on 3D CT synthesis

The dimensions of AP and LAT images were expanded by repeating images in the LAT and AP directions, respectively. Therefore, the inputs have identical spatial sizes as the target 3D sub-volume. Similarly, the superior-inferior coordinates of the sampled 3D CT sub-volume were incorporated as input by adding additional channels. We set the batch size to 1, and the network was trained for 170k steps. The data augmentation was not used. Training and testing were performed on a single DGX-A100 GPU with 40GB memory (NVIDIA Santa Clara, California, USA). Our code is available online (https://github.com/siyeopyoon/3D-Patch-Diffusion).

We quantitatively evaluated the performance of our 3D diffusion model using peak signal-to-noise ratio (PSNR) and structural similarity index (SSIM) in comparison to the original CT scans. The evaluation included comparisons with 2-view and 36-view filtered back projection reconstruction (FBP), as well as variations dependent on the number of function evaluations (NFE) for diffusion sampling. The reported metrics are the mean and standard deviation of 15 test cases.

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5. RESULTS

Representative images demonstrate the potential of our 3D diffusion model to synthesize high-resolution wrist CT scans from AP and LAT image inputs. The 3D volume rendering and slices of the synthesized CT scans show anatomical agreement with the original CT images, while variations in trabecular bone structure are observed (**Fig.2**). The quantitative evaluation showed that CT generated with the diffusion model was superior in PSNR and SSIM compared to FBP CT reconstructed with 2 views (**Table 1**). In the comparison of the performance based on NFE, synthesized CTs with over 50 NFE had higher SSIM than FBP CTs with 36 views.

6. DISCUSSION AND CONCLUSION

We present a 3D diffusion model for synthesizing high-resolution 3D CT images from bidirectional X-ray images. Our diffusion model is conditioned on images and their corresponding patch coordinates. Therefore, the denoiser 3D U-net of the diffusion model can be trained using a patch loss function, which was executed on a single GPU. Further research is warranted to assess the performance of the 3D diffusion model in a group of patients with wrist fractures.

7. COMPLIANCE WITH ETHICAL STANDARDS

We obtained de-identified retrospective data for Mass General Brigham hospitals in compliance with HIPAA regulations and with the approval of the local institutional review board under protocol 2021P003404, which granted a waiver of consent.

8. ACKNOWLEDGMENTS

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